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A comprehensive review: surface modification strategies to enhance corrosion resistance of zirconia-based biomaterials in implant applications

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Abstract

In contemporary dental practice, there is a significant demand for materials that not only exhibit superior mechanical strength and durability but also offer excellent biocompatibility and aesthetic appeal. Zirconia (ZrO_2) has emerged as a leading biomaterial addressing these needs, owing to its exceptional properties such as high fracture toughness, resistance to corrosion and wear, and tooth-like translucency. These characteristics make zirconia ideal for various dental prosthetics, including crowns, bridges, and abutments. Advancements in zirconia composites, particularly yttria-stabilized zirconia (YSZ) and zirconia-toughened alumina, have further enhanced its mechanical properties and stability. YSZ, for instance, is widely utilized in dental ceramics due to its increased strength and fracture toughness. To optimize zirconia's performance, especially in terms of corrosion resistance and osseointegration, various surface modification techniques have been developed. These techniques encompass acid etching, coating, polishing, biofunctionalization, sandblasting, ultraviolet light exposure, and laser treatments. These modifications significantly improve bone integration by altering surface texture, structure, wettability, chemical resilience, and antibacterial characteristics. In summary, zirconia's combination of mechanical strength, biocompatibility, and aesthetic appeal, along with ongoing advancements in composite formulations and surface treatments, solidify its role as a leading biomaterial in implant applications.

Keywords Zirconia (ZrO_2), Surface functionalization, Bio-ceramic materials, Osseointegration, Aesthetic properties, Bioactive properties, Clinical performance, Bio-medical implants

Introduction

Ceramics, originating from the Greek term “*keramos*,” which means pottery or fired material, include substances such as glass, advanced ceramics, and cement-based materials. These inorganic, non-metallic solids are produced through heat treatment and cooling, forming covalent, ionic, or occasionally metallic bonds. Among these, zirconium dioxide (ZrO_2), or zirconia, is particularly significant in medical devices due to its superior properties (Bauer et al. 2013). Zirconium (Zr), a transition metal with the atomic number 40, is renowned for its exceptional mechanical, thermal, and corrosion-resistant

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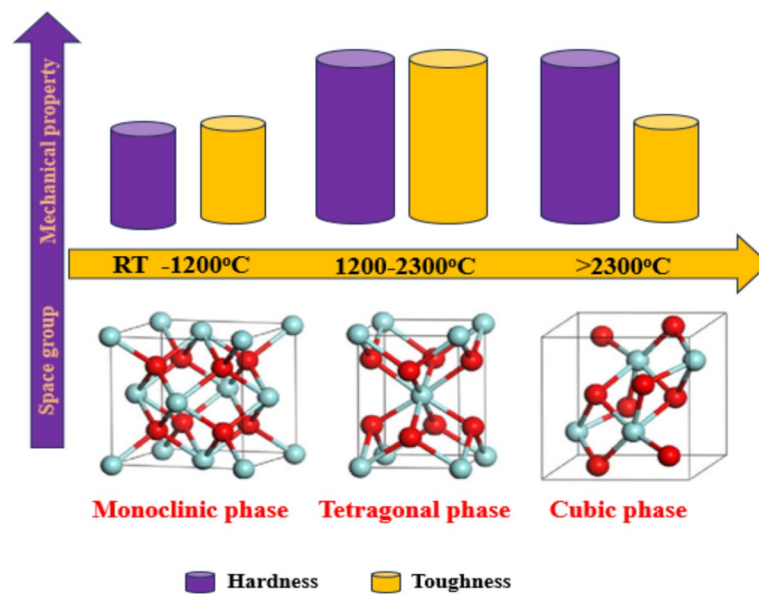


Fig. 1 A schematic illustration of the space groups for three distinct polymorphs of ZrO₂, along with their respective mechanical properties, adapted from reference Hannink et al. (2000) with permission

properties, making it highly suitable for various industrial and biomedical applications. Its most abundant isotope, ⁹⁰Zr, constitutes over half of naturally occurring zirconium. Zr is typically found in nature in the form of zirconia (ZrO₂), primarily in the mineral baddeleyite. However, natural zirconia contains impurities, including radioactive and elemental elements, which must be removed through complex processes to obtain pure zirconia powder for biomedical use (Piconi et al. 1999; Deville et al. 2006; Kuboki et al. 1999). In dentistry, titanium and its alloys have traditionally been the materials of choice for implants due to their biocompatibility, mechanical strength, and clinical success (Deprich et al. 2008b, a; Steinemann et al. 1998; Heydecke et al. 1999). However, titanium's drawbacks, including its unaesthetic grayish color (Tschernitschek et al. 2005) susceptibility to galvanic reactions with saliva and fluoride (Sternner et al. 2004) and potential inflammatory effects caused by titanium particles (Akagawa et al. 1993), have driven the search for alternatives. Zirconia has emerged as a promising alternative, offering excellent osseointegration (Scarano et al. 2003; Kohal et al. 2004), superior biocompatibility (Steinemann et al. 1998), and aesthetic advantages due to its tooth-like color (Jackson et al. 1999). Additionally, zirconia is radiopaque and resists bacterial colonization more effectively than titanium (Ahmad et al. 1998; Pekkan et al. 2011; Rimodini et al. 2002; Gahlert et al. 2018). Unlike titanium, zirconia does not produce corrosion byproducts at the implant site (Sternner et al. 2004). These exceptional properties underscore zirconia's

potential as an outstanding material for dental implants. Its dense oxide layer provides long-term corrosion resistance, even in highly acidic environments (Qi et al. 2022). In addition, zirconia's excellent wear resistance and minimal metal ion release help to mitigate the risks of inflammation and toxicity. This review emphasizes that while zirconia offers superior corrosion resistance—especially against pitting and crevice corrosion (Muller-Heupt et al. 2022)—its clinical limitations lie in brittleness, lower elastic modulus, and suboptimal osseointegration. To overcome these drawbacks, recent advances focus on material reinforcement (e.g., YSZ composites), surface modification (e.g., coatings, laser treatments, bioactive functionalization), and implant design innovations. These strategies aim to address zirconia's mechanical and biological limitations, ensuring improved performance and broader applicability in dental implantology.

Zirconia exists in three distinct crystalline forms

Zirconia exists in three distinct phases: monoclinic (stable at room temperature), tetragonal (stable above 1170 °C), and cubic (stable above 2370 °C) (Fig. 1). These phases vary in their coordination numbers, with the monoclinic phase being thermodynamically stable at lower temperatures due to the strong covalent Zr–O bonds. However, doping with oxides such as MgO, Y₂O₃, CaO, and Ce₂O₃ can stabilize the tetragonal or cubic phases even at room temperature, improving both the mechanical properties and phase stability of the material. The tetragonal phase, in particular, is prized for its

Table 1 Properties of various ceramics for biomedical applications (Chevalier et al. 1999; Dana et al. 2023)

Property	HAP	YTZP-A	Mg-PSZ	TZP	Alumina	Silicon carbide	Units
Chemical composition	$\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$	$\text{ZrO}_2/\text{Y}_2\text{O}_3/\text{Al}_2\text{O}_3$ 95/5/0.25	$\text{ZrO}_2 + 8\text{--}10 \text{ mol } \%$	$\text{ZrO}_2 + 3 \text{ mol } \%$	99.9% + MgO	SiC	
Density	3.4	6.05	5.7–6	> 6	≥ 3.97	3.2	g cm^{-3}
Porosity	-	0	-	< 0.1	< 0.1	-	%
Bending strength	600	1200	450–700	900–1200	> 500	-	MPa
Compression strength	1000	2000	2000	2000	4100	2000	MPa
Young's modulus	100–200	210	200	210	380	410	GPa
Fracture toughness	~3	12	7–15	7–10	4	-	$\text{MPa m}^{1/2}$
Thermal expansion coefficient	-	11×10^{-6}	$7\text{--}10 \times 10^{-6}$	11×10^{-6}	8×10^{-6}	4.3	K^{-1}
Thermal conductivity	-	2	2	2	30	8.4×10^{-6}	$\text{W m}^{-1} \text{K}^{-1}$
Hardness	350	1200	1200	1200	2200	-	HVN 0.1

enhanced stability and defect-induced activity. Moreover, it exhibits superior performance in catalytic applications (Kelly et al. 2008).

Categorization of biomedical zirconia

Magnesium partially stabilized zirconia (Mg-PSZ), yttrium-stabilized tetragonal zirconia polycrystals (3Y-TZP), and other materials like hydroxyapatite (HAP) and silicon carbide (SiC) are notable in biomedical applications. Mg-PSZ, a bi-phasic material comprising cubic grains and tetragonal precipitates with approximately 8 mol% MgO, has limited application in biomedicine due to its inadequate phase stability, high porosity, and suboptimal mechanical properties (Hannink et al. 2000; Pezzotti et al. 2014). On the other hand, 3Y-TZP, a monophasic zirconia variant stabilized with 2–3 mol% Y_2O_3 , demonstrates excellent strength and fracture toughness, making it highly suitable for therapeutic and dental uses (Hannink et al. 2000). Zirconia-toughened alumina (ZTA), a composite containing 5–30% zirconia within an alumina matrix, combines improved mechanical strength, wear resistance, fracture toughness, and resistance to aging degradation by optimizing the size and distribution of particles, without requiring additional dopants (Dongxu Tang et al. 2012) (Table 1).

Ceramic implants

Zirconia in dental implant applications

The introduction of the first ceramic dental implant dates back to 1974 with the development of the Tübingen implant, which was crafted from alumina (Al_2O_3) to overcome the limitations of metal implants Umar DM et al. (2024). Despite its biocompatibility and resistance to plaque, alumina implants were discontinued due to frequent fractures and low survival rates (Han J et al. 2017). Zirconia, with superior mechanical properties

compared to alumina, marked a significant advancement in ceramic implants (Andreietelli et al. 2009a, b) with the development of the Z-system® by Volz (Han J et al. 2017; Carinci et al. 2004). This innovation led to the application of zirconia ceramics in dental implantology (Fig. 2). Zirconia implants, introduced as a metal-free alternative, excel in aesthetics due to their white color, making them ideal for patients with thin gingiva or hypersensitivity to titanium. They show excellent biocompatibility, minimal immune reaction, and strong integration with both bone and soft tissues. Surface treatments such as acid etching, UV irradiation, and plasma treatments enhance their bioactivity and promote osteoblast attachment and osseointegration. One-piece zirconia implants reduce risks of bacterial colonization and peri-implantitis compared to two-piece designs, though their fabrication poses technical challenges due to ceramic brittleness. While survival rates for zirconia implants (74–98% after 12–56 months) are promising, they remain slightly lower than titanium implants (97–100% over 3–10.4 years), with issues like early failure, fractures, and bone loss requiring further improvements (Oliva et al. 2010; Depprich et al. 2008b, a; Akagawa et al. 1998). The corrosion resistance of zirconia in medical implants is a critical factor influencing their performance and longevity. Zirconia, particularly in its stabilized forms, exhibits excellent corrosion resistance due to the formation of a protective oxide layer, which enhances its biocompatibility and suitability for various biomedical applications (Carinci et al. 2004). This overview will delve into the corrosion behavior, advantages, and implications of zirconia in medical implants.

Advancements in zirconia ceramics for orthopedic implants

Early microscopic and radiographic studies in animals demonstrated the excellent biocompatibility of zirconia

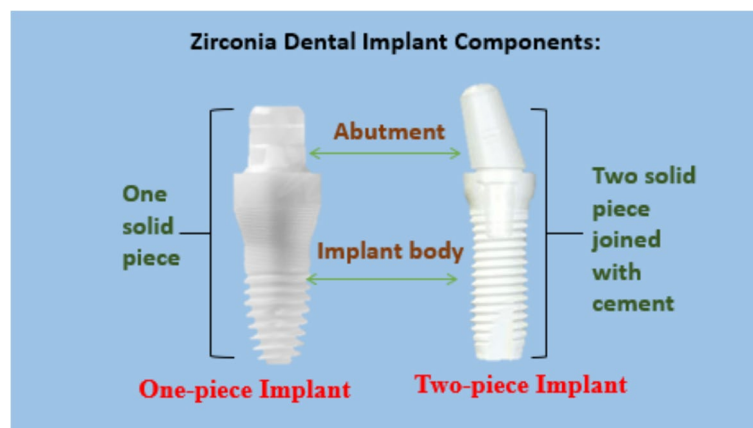


Fig. 2 Structural distinctions between one-piece and two-piece zirconia implants. **a** The first two components on the left represent zirconia abutments and implant bodies, which can be bonded to form a two-piece zirconia implant system. **b** The two components on the right showcase different configurations of one-piece zirconia implants, adapted from reference Yuriy May et al. (2025) with permission

implants, confirming that bone apposition could occur directly at the implant surface without gaps or cellular infiltration (Karthik et al. 2018). Zirconia bio-ceramics have been utilized in various biomedical applications, including orthopedic implants, thin films and coatings for metallic implants, porous bone scaffolds, substitute materials, and bone cements (Stadlinger et al. 2010). Despite extensive research, the impact of surface modifications on the biological response and mechanical stability of zirconia implants remained underexplored until recent advancements. Current efforts focus on optimizing zirconia surfaces to enhance their integration with bone tissue while improving mechanical and biological performance. Stadlinger et al. (2010) compared the bone-implant contact (BIC) and peri-implant bone volume density (rBVD) of one-piece zirconia and titanium implants in minipigs' mandibles. Zirconia implants were tested in submerged and non-submerged conditions, while titanium implants were all submerged. The results showed the following: BIC, submerged zirconia = 53%, submerged titanium = 53%, non-submerged zirconia = 48%; rBVD, submerged zirconia = 80%, submerged titanium = 74%, non-submerged zirconia = 63%. Zirconia implants exhibited comparable or superior results, particularly in rBVD (Gomes et al. 2011). Att et al. (2009) investigated the effect of UV treatment on zirconia (Y-PSZ) implants, finding enhanced osteoblast attachment, proliferation, and mineralization. UV treatment made the zirconia surface hydrophilic and reduced surface carbon, improving bioactivity. Han et al. (2011) later studied alumina-sandblasted yttrium-stabilized tetragonal zirconia polycrystals (Y-TZP) and found increased osteoblast differentiation and proliferation when combined with bone morphogenetic protein-2 (BMP-2). Lee

et al. (2013) also observed improved osseointegration and early bone formation around alumina-sandblasted Y-TZP implants with recombinant human BMP (rhBMP-2) gel. However, a recent study by Kohal et al. (2013) showed that sandblasted and acid-etched zirconia (TZP) implants exhibited slower cell proliferation and osseointegration compared to titanium implants, though they were still accepted by osteoblastic cells and supported osseointegration in rat bone tissue after 28 days.

Zirconia ceramic coatings and thin films for enhanced performance in orthopedic implants

Liu et al. (2006) utilized a filtered cathodic arc system with oxygen plasma to deposit zirconia thin films, producing stoichiometric zirconia nanoparticles that promoted bone-like apatite formation in simulated body fluid (SBF) due to their nanoscale surface features. These films demonstrated remarkable bioactivity and cytocompatibility, as evidenced by the growth and proliferation of bone marrow mesenchymal stem (BMMS) cells. Solazzo et al. (2008b, a) highlighted the potential of these zirconia coatings for enhancing the biocompatibility and osseointegration of metallic implants. Subsequent research confirmed that zirconia coatings improve the osseointegration properties of implants. Han et al. (2009) synthesized monoclinic zirconia thin films via micro-arc oxidation and exposed them to UV light, leading to the formation of hydrophilic groups on the surface, which enhanced bioactivity, as shown by increased apatite formation following UV treatment. G. Wang et al. (2010a, b) explored osteoblast behavior on nanostructured monoclinic zirconia coatings produced by plasma spraying. Their study demonstrated excellent in vitro bioactivity, with bone-like apatite formation on the coating surface

after 6 days in SBF. Cellular differentiation tests showed successful attachment, adhesion, and proliferation of MG63 osteoblast-like cells. Furthermore, zirconia was combined with bioactive hydroxyapatite (HAp) to create ZrO₂/HAp composite coatings for orthopedic implants, offering enhanced mechanical properties and biocompatibility compared to pure ZrO₂ or HAp. J. Wang et al. (2010a, b) further developed nanocomposite coatings on titanium through electrodeposition, incorporating fluorinated HAp and zirconia. These coatings exhibited strong bonding, low dissolution rates, and improved in vitro bioactivity.

Zirconia ceramic scaffolds and bone regeneration solutions: advancements in bone grafts and cements

Porous bone scaffolds play a critical role in bone tissue engineering, serving to fill defects and promote regeneration. Extensive research has been conducted on biodegradable polymers, ceramics, and polymer-ceramic nanocomposites with porous structures for these applications (Hutmacher et al. 2000; S. Yang et al. 2021). Zirconia, with its exceptional biomechanical properties, has emerged as a promising material for bone substitutes (S. Yang et al. 2002). Chen et al. (2006) developed zirconia-based porous scaffolds using replication and electrospray techniques, yielding scaffolds with superior mechanical strength and greater stiffness than those made using traditional slurry dip coating methods. This improvement was attributed to the reduction in microcracks and micropores, which enhanced the mechanical reliability of the zirconia scaffolds. When reinforced with nanocrystalline calcium phosphate (CaP) powders such as hydroxyapatite (HAp) and tricalcium phosphate (TCP), zirconia displays outstanding mechanical properties, including high bending and compressive strengths, stiffness, microhardness, and fracture toughness. HAp coatings and reinforcements also improve the biological activity and osseointegration of zirconia ceramics. As a result, significant attention has been given to the synthesis, characterization, sintering, and phase stability of zirconia-HAp composite scaffolds for bone regeneration. It has been noted that bone ingrowth is more prominent in microporous HAp scaffolds than in zirconia scaffolds with similar porosity. However, the mechanical toughness of zirconia complements HAp and TCP ceramics, providing additional strength. For example, Matsumoto et al. (2011) achieved impressive results with microporous zirconia/HAp composite scaffolds in a 70/30 ratio, demonstrating strength comparable to cortical bone, high protein adsorption, and tissue affinity shown in Fig. 3.

Further studies by the same team revealed that the compressive strength of zirconia/HAp scaffolds, with

porosity ranging from 72 to 91%, increased from 2.5 to 13.8 MPa by adjusting the zirconia content from 50% to 100 wt.%. Later developed porous scaffolds made from tetragonal zirconia, biphasic calcium phosphate, and poly(caprolactone) multilayers via extrusion. These scaffolds exhibited excellent cell viability, attachment, high porosity (78%), and compressive strength (12.7 MPa) (Mondal et al. 2013). Kotha et al. (2009) investigated the reinforcement of polymethylmethacrylate (PMMA) bone cement by incorporating high-strength zirconia fibers, both with and without an acrylic coating. The zirconia fibers, approximately 200 µm in length and with diameters of either 15 or 30 µm, were encapsulated in PMMA through a novel emulsion polymerization process. The study found that adding zirconia fibers enhanced the tensile and compressive properties, as well as the fracture toughness, of the bone cement. Specifically, fracture toughness increased by 23% with 15-µm fibers and by 41% with 30-µm fibers, while acrylic coating did not significantly improve the material properties. Based on these results, the study recommends using uncoated or acrylic-coated 30-µm zirconia fibers to significantly enhance the strength and fracture toughness of bone cements.

Additionally, Pattnaik et al. (2011) developed a multicomponent porous scaffold made from chitosan, nanoscale silica, and zirconia. These scaffolds demonstrated improved protein adsorption, biomineralization, and controlled biodegradation properties, and were non-toxic to rat osteoprogenitor cells, indicating their potential for bone tissue engineering applications. In another study, Gillani, Qiao, and Webster investigated the effect of incorporating zirconium dioxide (ZrO₂) nanoparticles into PMMA bone cement. The study showed that osteoblast adhesion was significantly higher on bone cements containing unfunctionalized nano-ZrO₂ compared to unmodified cements after 4 h. Furthermore, osteoblast proliferation was also significantly greater on ZrO₂-containing bone cements after 24 h, suggesting that the addition of ZrO₂ nanoparticles improves the bioactivity of bone cement, promoting better osteoblast adhesion and proliferation (Gillani et al. 2009). Lee et al. (2009) assessed the osseointegration of nano-technology-modified, micro-structured zirconia implant surfaces in rabbit femoral condyles.

The study compared calcium phosphate (CaP)-nano-coated zirconia implants (types A and C) with uncoated zirconia (ZiUnite) and titanium porous oxide (TiUnite) implants. After 3 weeks, TiUnite implants exhibited significantly greater bone-implant contact (BIC) (77.6%) compared to the CaP-coated zirconia implants A (64.6%) and C (62.2%). However, after 6 weeks, no significant differences in BIC were observed between the

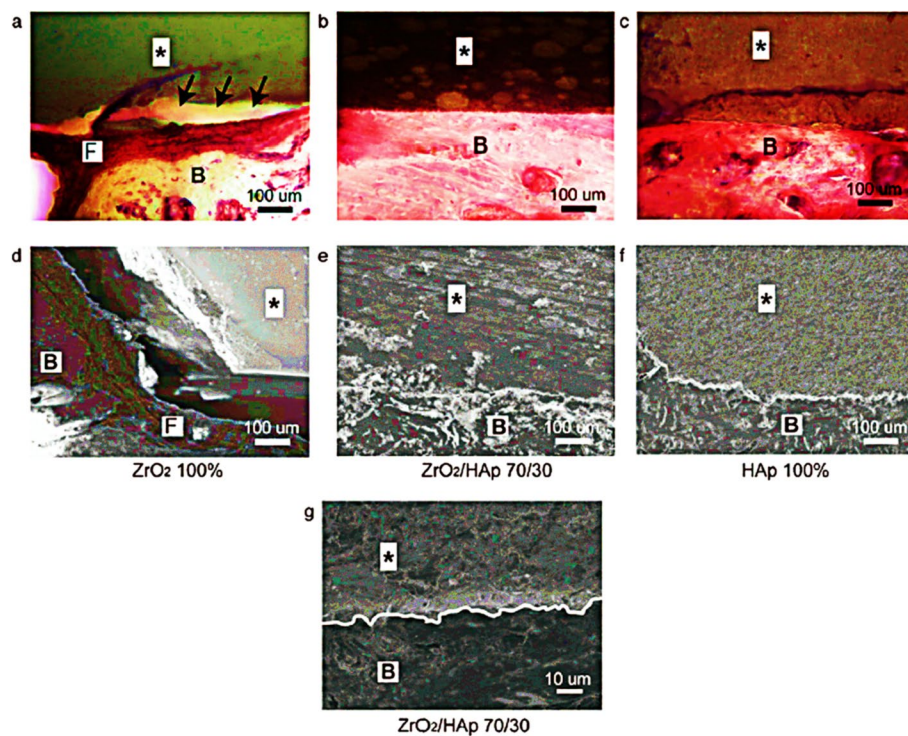


Fig. 3 Histological sections of the specimens are shown with HE-stained samples (a–c) and SEM images (d–g). The panels represent the following: a and d depict ZrO₂; b, e, and g illustrate ZrO₂/HAp-L; and c and f show HAp-L. Labels indicate * for the sample, B for bone, and F for fibrous tissue, adapted from reference Matsumoto et al. (2011) with permission

groups, concluding that while TiUnite and ZiUnite surfaces demonstrate high osseointegration, the addition of CaP nano-coating did not enhance the osteoconductivity of zirconia implant surfaces. Finally, Koller et al. (2011) studied the survival rates of teeth, implants, and double crown-retained removable dental prostheses (RDPs). The study revealed that the survival rates of teeth ranged from 60.6 to 95.3% over periods of 4 to 10 years. RDPs supported by teeth had survival rates between 90.0 and 95.1% after 4 to 5.3 years. Implants supporting prostheses in the mandible showed survival rates between 97 and 100% over 3 to 10.4 years, and implant-retained RDPs in the mandible had survival rates between 95 and 100% after 9 to 10.4 years. The study concluded that the long-term outcomes of double crown-retained RDPs remain under-researched, recommending further studies with higher levels of evidence to confirm these treatment modalities.

A comparative analysis of zirconia and titanium implants

This section presents a concise comparison between zirconia and titanium implants, highlighting their key differences and similarities. The analysis focuses on critical factors such as material properties, cost, aesthetics, strength, biocompatibility, and clinical performance. By

organizing the data into a structured table, it enhances clarity and supports informed decision-making for clinical applications. The comparison also identifies current limitations and ongoing research directions for zirconia implants (Table 2).

The choice between zirconia and titanium implants depends on their distinct benefits and limitations. Titanium offers proven long-term reliability, while zirconia appeals to patients seeking superior aesthetics. A dentist's consultation is crucial to select the most appropriate option based on individual needs and clinical conditions GS Kaliaraj et al. (2016).

Corrosion resistance of zirconia implants

A study (Subramanian et al. 2015) assessed zirconia and zirconia-silica coatings on titanium dental implants to enhance corrosion resistance. Single-layer coatings, unlike double or triple layers with structural flaws, showed reduced surface roughness and improved corrosion resistance (Fig. 6).

Zirconia-silica layers provided superior protection and demonstrated stability in artificial saliva over 3 weeks. Biocompatibility tests confirmed osteoblast compatibility, highlighting single zirconia-silica coatings as promising for titanium implants (Table 3).

Table 2 Comparative properties of zirconia and titanium dental implants

Category	Zirconia implants	Titanium implants	Ref
Cost	\$500–\$600 per implant; More expensive	\$300–\$500per implant; total cost gener- ally lower	(Oliva et al. 2010)
Aesthetics	Superior aesthetics; eliminates gray lines in thin gums	May cause visible gray lines; ceramic abutments can reduce this issue (Fig. 4)	
Strength	Durable but less flexible, prone to micro- fractures	High strength, flexibility, and superior fracture resistance	
Biocompatibility	Lower inflammatory, prominent better in soft tissue engineering	High biocompatibility but may induce a stronger inflammatory response	
Osseointegration	Promising but lower bioactivity and inte- gration rates	Well-established direct bone integration with high success rates (Fig.5)	(Andreiotelli et al. 2009a, b)
Surface modifications	Various bioactive and therapeutic treat- ments to improve osseointegration	Already optimized for osseointegration, minimal modifications needed	(Depprich et al. 2008b, a; Dlugon et al. 2017)
Clinical applications	Still under investigation; ongoing research needed	Long history of clinical success and reli- ability	(Andreiotelli et al. 2009a, b; Ito et al. 1995)
Ease of placement	Single-piece design is harder to place; newer two-piece models offer flexibility but need more testing	Easier to place, even in complex cases; well-established procedure	(Dlugon et al. 2017)



Fig. 4 Comparison of the visual appeal between titanium and zirconia abutments in dental implants, adapted from reference Oliva et al. (2010) with permission

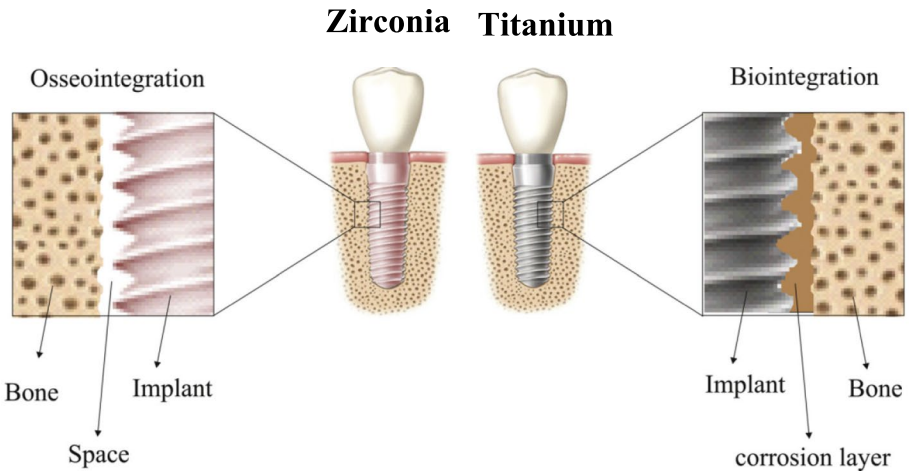


Fig. 5 Osseointegration versus bio-integration, adapted from reference Andreiotelli et al. (2009a, b) with permission

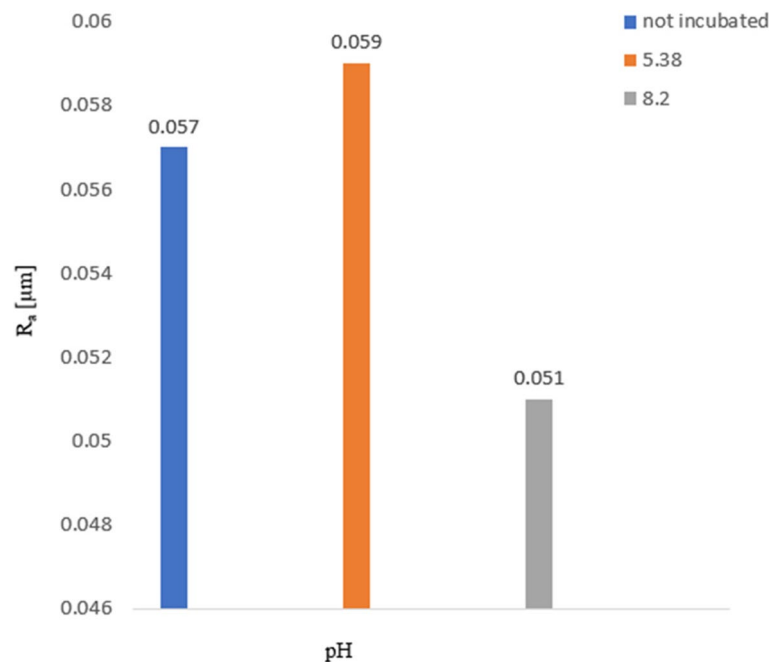


Fig. 6 Surface roughness parameters of titanium samples coated with a single zirconia-silica layer, both before and after incubation in acidic and alkaline artificial saliva solutions. Adapted from reference Subramanian et al. (2015) with permission

Table 3 Corrosion characteristics of titanium without any coating and after the application of sol-gel coatings (Subramanian et al. 2015)

Sample	$E_{COR}(mV)$	$E_{CO}(mV)$	$ βa (mV \text{ dec}^{-1})$	$ βc (mV \text{ dec}^{-1})$	$R_p(MΩ \text{ cm}^2)$	$J_{COR}(nA \text{ cm}^2)$
Ref Ti	− 378.0	− 380.6	189	185	0.34	120.0
Zr-Ti	223.2	− 230.2	165	142	12.70	2.61
Zr-Si-Ti	− 43.8	− 37.9	285	182	17.15	6.77

Potentiodynamic polarization studies revealed that coatings on titanium greatly enhanced corrosion resistance in Ringer's solution after 1 h of immersion. The corrosion potential shifted positively, and the current density decreased by approximately two orders of magnitude. Among the coatings, 1Si-Zr-Ti demonstrated notable changes in Tafel slopes, with anodic reactions being predominant. This was attributed to the minimized contact between titanium and the electrolyte, which increased charge-transfer resistance and slowed down the corrosion process (Fig. 7).

New titanium alloys, such as Ti-Zr-Pd and Ti-Sn-Pd, exhibit superior cytocompatibility compared to Ti-6 Al-4 V ELI, with Ti-15Zr4 Nb2 Ta-0.2Pd and Ti-15Sn4 Nb2 Ta-0.2Pd showing the best performance in L929 and MC3 T3-E1 cell tests. These alloys also demonstrate enhanced tensile strength, with Ti-15Sn4 Nb2 Ta-0.2Pd achieving a strength of 989 MPa and elongation of 14.4%, and Ti-15Zr4 Nb2 Ta-0.2Pd showing 725 MPa strength

and 23.6% elongation. However, zirconium or tin content above 20% can decrease ductility, indicating the need for balanced compositions (Fig. 8). The new alloys also exhibit improved corrosion resistance. Titanium release into hydrochloric acid solution was measured at 1300, 20–50, 7.5–13, and 20–25 $\text{pg/cm}^2/\text{day}$ for Ti-6 Al-4 V ELI, Ti-Zr, Ti-Zr-Pd, and Ti-Sn-Pd alloys, respectively. In phosphate-buffered saline (PBS), titanium release was much lower, with Ti-6 Al-4 V ELI and Ti-15Sn4 Nb2 Ta-0.2Pd releasing 2 $\text{pg/cm}^2/\text{day}$, and Ti-15Zr4 Nb2 Ta-0.2Pd releasing just 1 $\text{pg/cm}^2/\text{day}$. Corrosion resistance followed the order Ti-Zr-Pd > Ti-Sn-Pd > Ti-Zr > Ti-6 Al-4 V ELI, consistent with the cytocompatibility results (Fig. 9), although a reversal in performance was observed between Ti-Zr and Ti-6 Al-4 V ELI in the L929 cell extraction tests. Cytocompatibility of Ti-Zr, Ti-Zr-Pd, and Ti-Sn-Pd alloys is shown, with error bars indicating standard deviations (J. Pan et al. 1997).

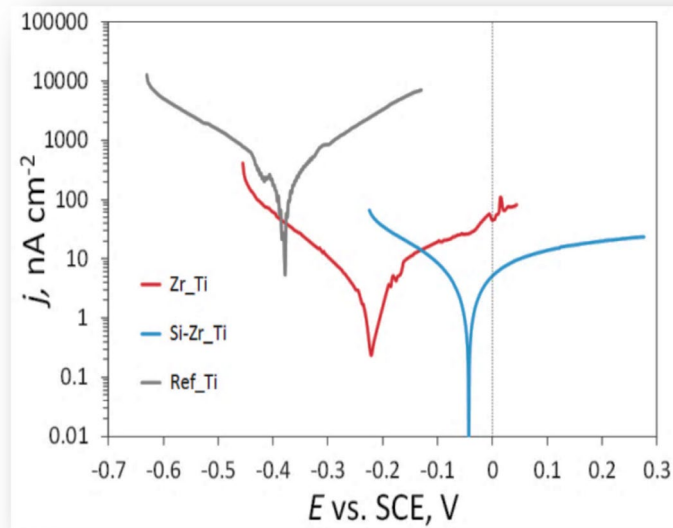


Fig. 7 Potentiodynamic polarization curves were obtained for the untreated titanium surface (Ref. Ti) as well as for titanium samples coated using the sol-gel technique (Zr-Ti and Si-Zr-Ti). Adapted from reference Subramanian et al. (2015) with permission

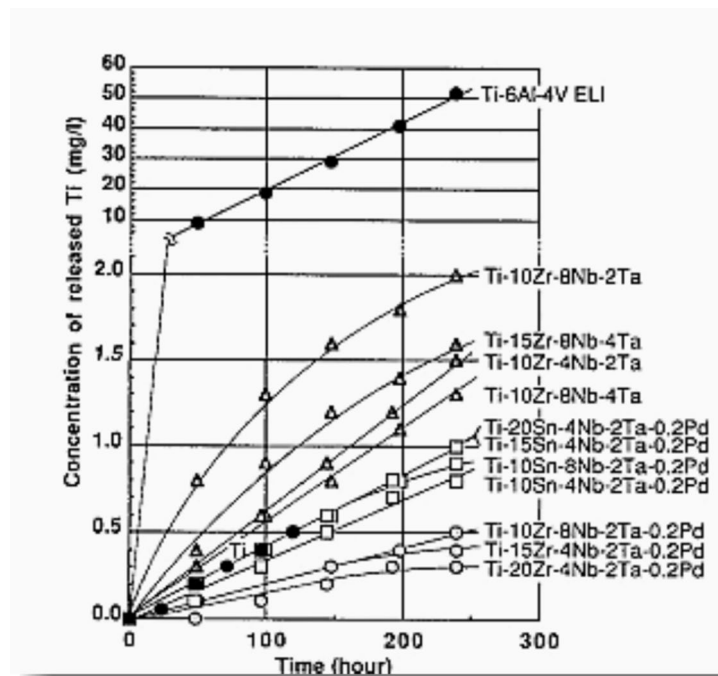


Fig. 8 A comparative analysis of titanium alloy compositions and their ion release behavior in corrosive environments over time. Adapted from reference J. Pan et al. (1997) with permission

Zirconium oxide (ZrO_2) coatings demonstrated excellent biocompatibility with fibroblast cells, exhibiting higher viability and better adhesion compared to

bare 316L stainless steel (SS). Fluorescence microscopy showed healthy cell morphology on the ZrO_2 surface, likely due to its oxide-rich composition, which supports

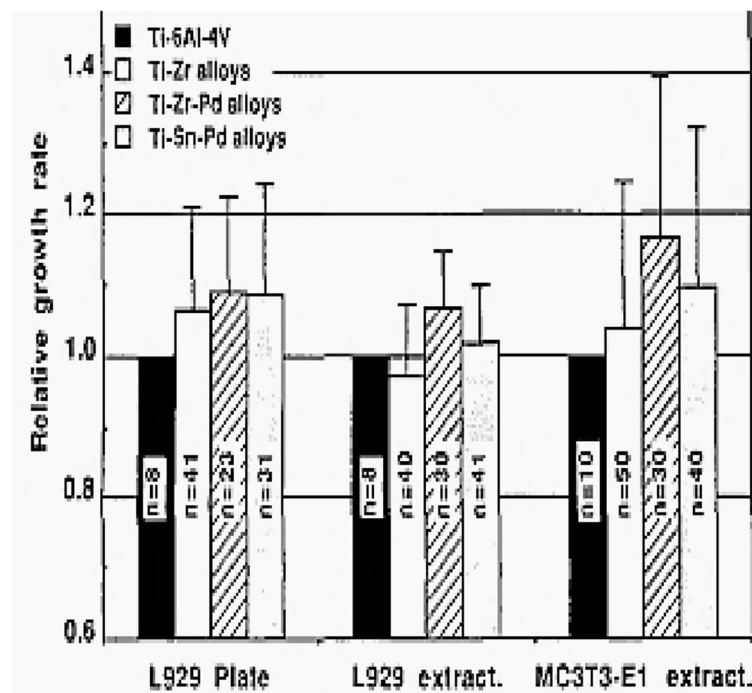


Fig. 9 A comparative evaluation of the relative growth rates of titanium-based alloys in various extracts. Adapted from reference J. Pan et al. (1997) with permission

cell attachment and differentiation. Electrochemical testing indicated robust corrosion resistance in acidic environments, with ZrO_2 maintaining a stable passive layer. Although exposure to fluoride ions in acidulated AS solutions (0.2% and 2% NaF) slightly weakened the coatings by forming zirconium fluoride (ZrF_4), ZrO_2 still showed strong passive behavior. In contrast, bare 316L SS experienced considerable corrosion due to fluoride-induced formation of soluble products. These findings position ZrO_2 coatings as effective in improving both biocompatibility and corrosion resistance in harsh conditions (Frateur et al. 2000; Satendra Kumar et al. 2010; Alves et al. 2009; Huang et al. 2003). In addition, zirconium-based alloys and grade 2 titanium surfaces supported cell viability, though at lower rates than plastic. Osteogenic induction resulted in reduced viability on all surfaces, but it did not significantly affect uniformity, suggesting that surface properties had minimal impact on cell survival. All alloys, including Zr97.5 Nb1.5Mo1.0 and Zr97.5 Nb1.5 Ta1.0, supported matrix mineralization, as shown by Alizarin Red staining and enhanced alkaline phosphatase (ALP) activity, with Zr97.5 Nb1.5Mo1.0 displaying the highest ALP levels. Gene expression analysis showed elevated levels of osteogenic markers, including osteocalcin (OC), osteopontin (OP), bone sialoprotein (BSP), and RUNX2 after induction. The alloys Zr97.5 Nb1.5Mo1.0 and Zr97.5 Nb1.5 W1.0 exhibited the highest RUNX2

expression, suggesting their potential to promote osteogenic differentiation. These findings support the use of these alloys for applications where bone integration is crucial Shyti G et al. (2014). Though Prawoto et al. (2012) focused on stainless steel, the study offers transferable insights for zirconia's behavior in biomedical environments. Its use of sodium hydroxide parallels zirconia's vulnerability to hydrothermal degradation under alkaline conditions. The mechanisms of crack initiation and stress-induced propagation discussed can inform strategies to manage similar issues in zirconia. Moreover, the emphasis on fracture morphology and stress redistribution aids in refining corrosion testing and durability assessments for zirconia-based implants. These insights are crucial as they help bridge knowledge from well-studied metallic systems to ceramic-based biomaterials like zirconia, guiding the development of more accurate predictive models and enhanced clinical performance.

Long-term corrosion resistance: zirconia vs titanium implants

Consider the famously tragic sinking of the Titanic, a ship primarily constructed from steel, an alloy of metal, which after decades at the bottom of the ocean, has mostly corroded and decayed into rust, dust, and remnants of marine life. Now, think of the ancient Greek and Roman amphorae-ceramic vessels originally used by the



Fig. 10 Comparative longevity of ceramic and metal: insights from ancient artifacts and modern implants, adapted from reference Yuriy May et al. (2025)

Etruscans to transport goods like wine. These amphorae were made from clay, particularly terracotta, and were fired at high temperatures to harden and seal them. While the Titanic, a metal vessel, deteriorated rapidly after sinking in 1912 and was nearly completely corroded by the 1980 s, many of these ancient ceramic containers have been recovered from the ocean floor after thousands of years and remain largely intact. Some even still contain liquid. Unlike metals, ceramics like terracotta do not corrode or rust over time, even in harsh, underwater environments. This durability contrasts sharply with materials like titanium alloys used in dental implants, which, though resistant to corrosion in many settings, may not provide the same level of longevity in the long term. Just as ceramic amphorae have withstood the test of time in a corrosive environment, ceramics in dental implants could offer a more resilient and long-lasting option for use in the mouth, a similarly wet and harsh environment, compared to metal alternatives that may degrade over time (Yuriy May et al. 2025) (Fig. 10). While amphorae are durable in passive marine environments, dental implants are subject to additional stresses such as mechanical loading, chemical interactions with saliva and food, and microbial activity, which may affect their longevity differently. The analogy may oversimplify the complexity of material behavior in biomedical applications.

Zirconia: a durable and biocompatible material for implant applications

Dental implants have gained significant popularity as a method for replacing missing teeth. In the USA, their usage has risen from 0.7% in 1999–2000 to 5.7% in 2015–2016, with projections suggesting that the prevalence

could reach 23% by 2026 Elani HW et al. (2018). According to Howe et al. (2019), titanium remains the material of choice for implants due to its high survival rate: 96.4% over 10 years. Sicilia et al. (2008) and Ioannidis et al. (2017) have discussed that titanium has limitations, such as allergic reactions, titanium deposits, mucosal discoloration, and aesthetic concerns. Andreiotelli et al. (2009a, b) have led to a search for alternative materials. Aluminum oxide was the first ceramic implant material, showing osseointegration but lacking in fracture toughness. This led to long-term survival rates of 65–92%. Despite its failure, it paved the way for the development of metal-free materials like zirconia, which has been used in dental implants since 1995 (Cionca et al. 2017). Zirconia is favored for its excellent mechanical properties, biocompatibility, and aesthetic appearance, though its fracture toughness requires further improvement (Glauser et al. 2004). Surface modifications, including morphology, chemical composition, and roughness, play a crucial role in enhancing tissue integration (Ko et al. 2007; R Teimouri et al. (2023)). Such modifications have been widely used in titanium implants and are now being explored for zirconia implants to improve biological performance and osseointegration studied in detail by Turon-Vinas et al. (2018), Deligianni et al. (2001), Bosshardt et al. (2017), and Han et al. (2020). Various surface modification methods for zirconia, including physical treatments, chemical treatments, and coatings, highlighting current advancements and future directions were discussed in detail.

The Zr97.5 Nb1.5 Ta1.0 alloy demonstrated exceptional corrosion resistance, indicated by high impedance values, along with excellent biocompatibility and osteogenic

potential. Although all surfaces supported comparable cell growth, Zr97.5 Nb1.5Mo1.0 and Zr97.5 Nb1.5 Ta1.0 stood out for their superior matrix mineralization and alkaline phosphatase activity. Zr97.5 Nb1.5Mo1.0 exhibited higher levels of proliferation markers, including podocalyxin and CD49f, following osteoinduction, while Zr97.5 Nb1.5 Ta1.0 fostered a more differentiated cell population, as indicated by increased mineralization and osteogenic marker expression. These characteristics position Zr97.5 Nb1.5 Ta1.0 as an excellent candidate for implant applications, offering both superior corrosion resistance and strong bone integration support. The study evaluated the differentiation potential of human BMSCs by analyzing proliferation markers (CD49f, podocalyxin) and osteogenic markers (Col 1, CD146, OC, OP, BSP, RUNX2). Zr-based alloys significantly upregulated osteogenic genes compared to controls. Zr97.5 Nb1.5Mo1.0 showed high podocalyxin and Col. 1 expression, while Zr97.5 Nb1.5 Ta1.0 had lower podocalyxin but supported osteogenic differentiation with moderate BSP and OP expression. Both alloys showed notable RUNX2 increases, indicating strong osteogenic potential, while grade 2 Ti showed minimal osteo-inductive response (Gautam et al. 2016; Shyti G et al. 2014; He et al. 2010; Rouwkema et al. 2006) (Fig. 11).

Zirconium oxide and nitride coatings on medical implants enhance wear resistance and corrosion protection, extending implant lifespan and reducing the need for revision surgeries. These coatings minimize corrosion and wear-related failures, leading to fewer complications and surgeries. Patients experience better outcomes, including reduced pain, improved mobility, and fewer interventions, which boosts their quality of life. The increased durability also lowers healthcare costs by reducing expenses related to revisions, rehabilitation, and hospital stays, making these coatings a cost-effective improvement in orthopedic care. Sacchetti et al. (2007) discovered a subset of self-renewing osteoprogenitor cells in human bone marrow sinusoids that are essential for creating the hematopoietic microenvironment (HME). These cells, marked by high MCAM/CD146 expression, can transfer the HME to other sites when transplanted, leading to the formation of similar subendothelial cells within a small bone organ. The development of these stromal cells in the bone marrow occurs through dynamic interactions with growing sinusoids. These subendothelial stromal cells, located along the sinusoidal wall, are key producers of Angiopoietin-1, a crucial factor for vascular remodeling and the formation of the hematopoietic stem cell (HSC) niche. The study underscores the interconnections between HME establishment, the presence of skeletal progenitors in bone marrow sinusoids, and angiogenesis. Lamour et al. (2007) investigated

the regulation of high bone sialoprotein (BSP) expression during human osteoblast differentiation. They discovered that the transcription factor Runx2 and histone deacetylase 3 (HDAC3) collaborate to repress BSP gene expression in undifferentiated osteoblasts. As osteoblasts differentiate, this repression is lifted, leading to increased BSP expression. The study highlights the dynamic interplay between transcription factors and epigenetic modifiers in controlling osteoblast-specific gene expression. The patent US5496359 A, titled "Zirconium oxide and nitride coated biocompatible leads," was granted to Smith and Nephew on March 5, 1996. This patent describes a method for coating medical devices, such as endoprostheses, with zirconium oxide or zirconium nitride to enhance their biocompatibility (DavidsonSmith JA et al. (1996). The process of coating medical devices with zirconium oxide (ZrO_2) or zirconium nitride enhances their interaction with biological tissues, thereby reducing the risk of negative reactions. This advancement is significant for the development of medical implants that offer better tissue compatibility and extended longevity. In a study by Gahlert et al. (2012), the osseointegration of zirconia (ZrO_2) and titanium (Ti) implants was compared in the maxillae of mini pigs. Both types of implants demonstrated direct bone-to-implant contact (BIC) and successful peri-implant bone density formation. Specifically, ZrO_2 implants had mean peri-implant bone density values of 60.4%, 65.4%, and 63.3% at 4, 8, and 12 weeks post-implantation, respectively. Titanium SLA (Ti-SLA) implants showed values of 61.1%, 63.6%, and 68.2% during the same intervals. The BIC ratio for ZrO_2 implants ranged from 67.1 to 70%, while Ti-SLA implants ranged from 64.7 to 83.7%. No significant differences were found between the two implant types at any of the measured time points. The results indicated that ZrO_2 implants achieved osseointegration similar to Ti-SLA implants, highlighting their potential as effective alternatives in dental implant procedures.

Özkurt and Kazazoğlu et al. (2011) reviewed studies on zirconia dental implants published from 1975 to 2009, finding that zirconia implants offer osseointegration comparable to titanium implants. They also noted that zirconia implants exhibit evenly distributed stress and similar surface roughness to titanium, both of which are important for implant stability and integration. Pieralli et al. (2018) evaluated the osseointegration of zirconia dental implants in animal studies using metrics like bone-to-implant contact (BIC), removal torque (RTQ), and push-in tests. The results showed no significant differences based on the bulk materials used, but highlighted variability across animal models, emphasizing the need for careful model selection in future studies. Zirconia ceramics, especially 3Y-TZP, are gaining popularity in

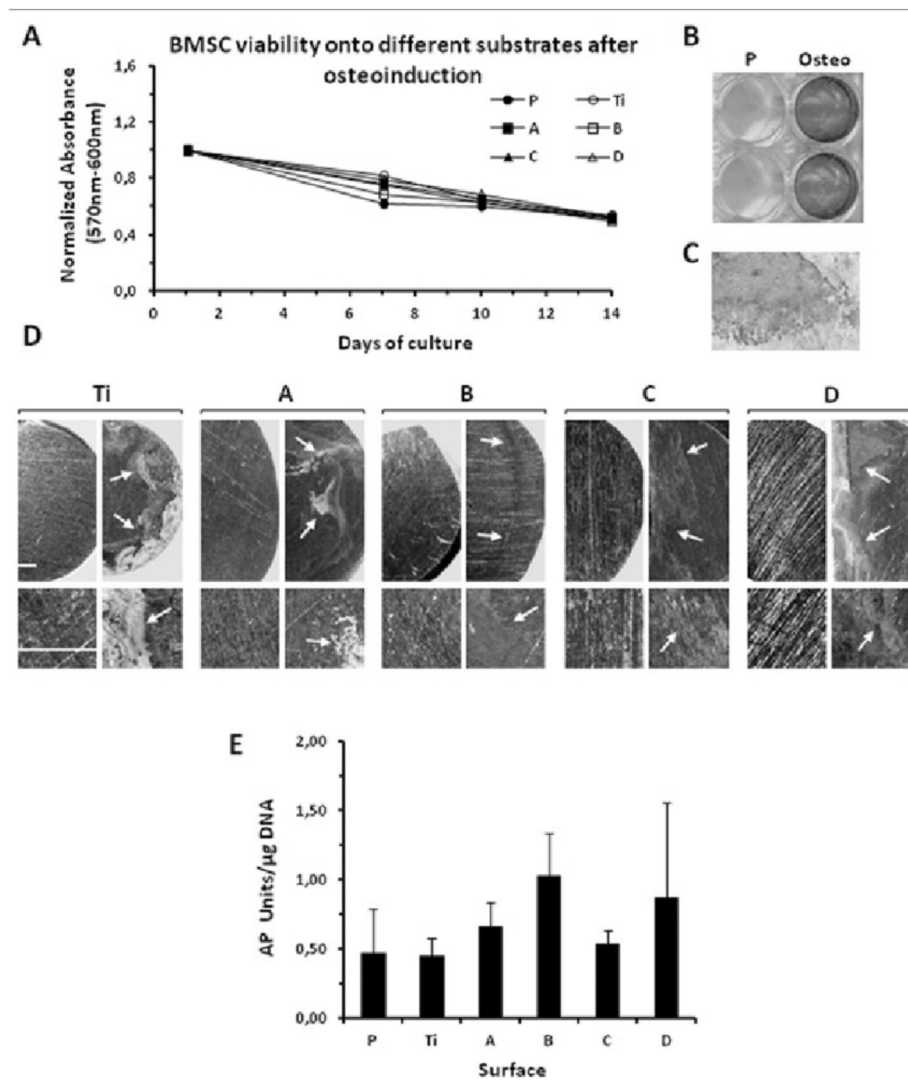


Fig. 11 Panel A: Growth kinetics of BMSC cells, with data normalized to the OD readings of control samples on plastic (P). Curves represent growth on various surfaces as described in the legend: P, plastic; Ti, Grade 2 Ti; A, Zr97.5 Nb1.5 Ti1.0; B, Zr97.5 Nb1.5 Mo1.0; C, Zr97.5 Nb1.5 W1.0; and D, Zr97.5 Nb1.5 Ta1.0. Data points represent the average of three measurements from two independent experiments (SD variations omitted for clarity). Panel B: Alizarin Red staining of mineralized matrix deposition in a representative BMSC cultured on plastic in standard medium (P) or with osteogenic induction (osteo). Panel C: Higher magnification image of one osteo sample from panel B. Panel D: Alizarin Red staining of Ti and alloy disks used as substrates for cell growth. Images show the metals before (left) and after cell seeding and osteogenic induction (right). Ti, A, B, C, and D as in panel A. Each metal shows low-magnification images in the top panels and higher magnification in the bottom panels, with white bars (only in Ti) at 0.5 mm. Arrows indicate mineralized matrix areas. Panel E: Alkaline phosphatase activity in BMSC on various substrates. Data are normalized to DNA content; bars represent the mean $\pm$ SE of three measurements from two independent experiments. Ti, A, B, C, and D as in panel A; P, plastic, adapted from reference Deligianni et al. (2001) with permissions

dental implants due to their strength, chemical stability, and biocompatibility, offering a metal-free alternative to titanium. While zirconia addresses issues like hypersensitivity and aesthetics, its reliability is currently lower than titanium, with survival rates ranging from 74 to 98% over 12–56 months and a 4.1% fracture rate within the first year, often during torquing. Marginal bone loss averages 1.31 mm for single-tooth and 1.95 mm for three-unit

prostheses after 1 year. Despite these challenges, zirconia abutments show good soft tissue integration and reduced bacterial colonization. The paper calls for improved manufacturing processes and more clinical data to boost zirconia implant reliability and adoption (Mohseni et al. 2023). Hoffmann et al. (2008) conducted a histological evaluation of bone apposition around zirconia dental implants (Z-system, Konstanz, Germany) at 2 and 4

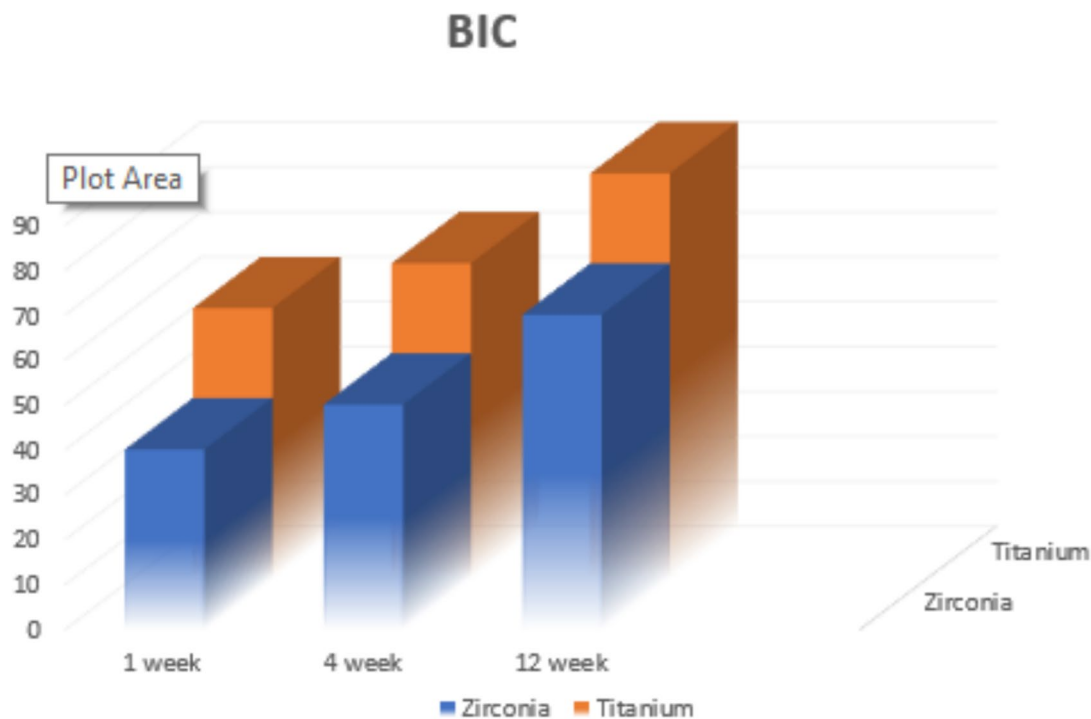


Fig. 12 A graph illustrating the progression of bone-to-implant contact (BIC) over time (1, 4, and 12 weeks) shows no significant difference between the two treatment groups ($p < 0.05$). Adapted from reference Rita Depprich et al. (2008b, a) with permission

weeks after implantation. At the 2-week mark, zirconia implants exhibited slightly greater bone apposition (54–55%) compared to titanium implants (42–52%). However, by the 4-week mark, titanium implants demonstrated greater bone apposition (68–91%) than zirconia implants (62–80%). Schultze-Mosgau et al. (2000) assessed bone-to-implant contact (BIC) in zirconia endodontic cones following apicectomy, revealing a BIC ratio of 0.95 for titanium surfaces and 1.47 for zirconia, indicating significantly better bone healing on zirconia surfaces compared to titanium. Kohal et al. (2004) compared the soft and hard tissue responses of sandblasted zirconia implants (re-implant) with sandblasted and acid-etched (SLA) titanium implants. After 9 months of healing and 5 months of loading, the mineralized BIC was 72.9% for titanium implants and 67.4% for zirconia implants. Schünemann et al. (2019) provide a comprehensive review on zirconia surface modifications aimed at enhancing its application in implant dentistry. Zirconia, due to its excellent biocompatibility, mechanical strength, and aesthetic appeal, is considered a strong candidate for dental implants. However, its bioinert nature necessitates surface modification to improve osseointegration. The authors discuss various techniques such as sandblasting, acid etching, laser treatment, and bioactive coatings, which can enhance surface roughness, chemical properties, and cellular responses. These modifications are shown

to promote better cell adhesion, proliferation, and bone-implant integration, thus improving the overall clinical performance of zirconia implants.

A separate study comparing BIC between zirconia and titanium implants over a 12-week period (Fig. 12) found no significant differences between the two materials. At 1 week, the mean BIC was $35.3\% \pm 10.8$ for zirconia and $47.7\% \pm 9.1$ for titanium. After 4 weeks (Fig. 13), the BIC values increased to $45.3\% \pm 15.7$ for zirconia and $58.6\% \pm 9.5$ for titanium. By 12 weeks (Fig. 14), BIC had further increased to $71.4\% \pm 17.8$ for zirconia and $82.9\% \pm 10.7$ for titanium implants. These results suggest that both zirconia and titanium implants achieve comparable osseointegration over time. Histological examination at 4 weeks showed mature regeneration tissue with osteoid and woven bone formation around both types of implants. At 12 weeks, mature lamellar bone was in direct contact with both zirconia and titanium surfaces, indicating successful integration with the hard tissue. No inflammation was observed in any of the specimens throughout the study (D Morena et al. 2024).

The findings support the use of zirconia implants as a viable alternative to titanium implants, as they demonstrate comparable osseointegration properties. However, it is crucial to acknowledge that while zirconia offers aesthetic benefits due to its tooth-like appearance, it may not provide the same level of strength and durability as

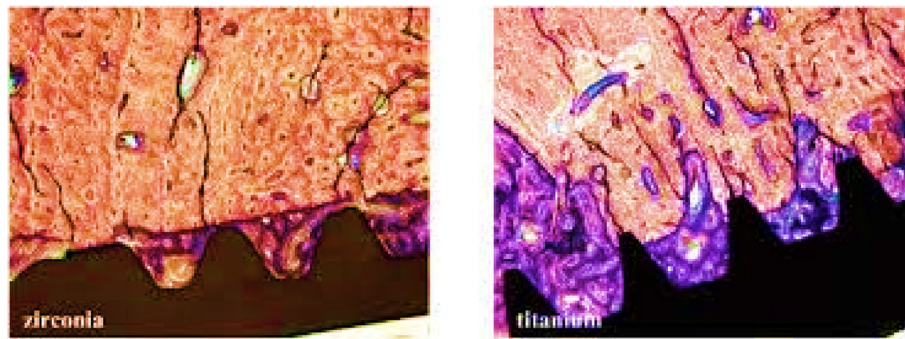


Fig. 13 At 4 weeks post-implantation, both zirconia (left) and titanium (right) implant surfaces displayed the formation of osteoid and woven bone (toluidine blue, $\times 50$ magnification). Adapted from reference Rita Depprich et al. (2008b, a) with permission

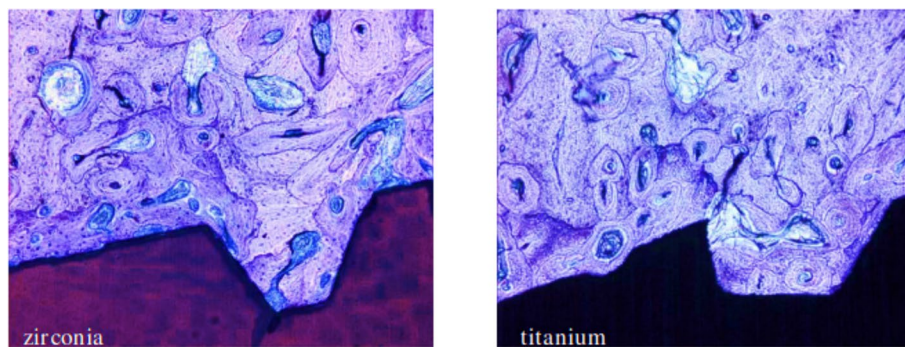


Fig. 14 After 12 weeks of healing, well-formed lamellar bone is observed in close proximity to both the zirconia implant (on the left) and the titanium implant (on the right) (toluidine blue, $\times 100$ magnification). Adapted from reference Rita Depprich et al. (2008b, a) with permission

titanium. Titanium implants have a well-established history of success and longevity, with higher survival rates and reduced marginal bone loss after 1 year (Rita Depprich et al. 2008b, a). Schultze-Mosgau et al. (2000) examined the osseointegration of zirconium oxide (ZrO_2) ceramic cones compared to titanium cones during apicectomy procedures. The study involved the insertion of 20 ZrO_2 cones and 20 titanium cones into the mandibles of Göttinger minipigs. Over a 6-month healing period, the researchers performed intravital polychrome sequence marking and conducted both qualitative light microscopy and fluorescence microscopy, as well as quantitative histomorphometric analysis. The results showed direct bone contact with both ZrO_2 and titanium surfaces, with bone remodeling processes occurring adjacent to both. Quantitatively, the mean ratio of total cone/bone contact to total cone/fibrous tissue contact was 0.95 on the titanium surface and 1.47 on the ZrO_2 surface, revealing a statistically significant difference ($p = 0.02$). The study concluded that the biocompatibility of ZrO_2 was on par with that of titanium, suggesting that ZrO_2 cones could be suitable for sealing resected teeth after apicectomy. As such, the choice between zirconia and titanium implants should be tailored to the

patient's specific needs, taking into account factors like aesthetics, strength requirements, and potential material sensitivities.

Surface modifications

Zirconia as a coating material

Sollazo et al. (2008) reported that zirconia-coated titanium implants exhibited a higher bone-to-implant contact (BIC) of 43.8% at 4 weeks, compared to 31.8% for titanium implants without coating, indicating that zirconia coatings can enhance osseointegration. In a separate study, Kirsten et al. (2015) assessed the bioactivity of a new glass composition (PC-XG3) zirconia coating through the simulated body fluid (SBF) immersion test. Both smooth and microstructured glass coatings on zirconia implants demonstrated strong adhesion to the substrate and significantly improved bioactivity following immersion in SBF (SM Jusoh, et al. 2018). Silicon carbide (SiC) nanocomposite coatings offer excellent corrosion resistance due to their chemical inertness, thermal stability, and mechanical strength. Acting as durable barriers, they reduce corrosion rates and prevent microcrack formation in harsh environments. When integrated into metal matrices like electroless nickel or enamel, SiC

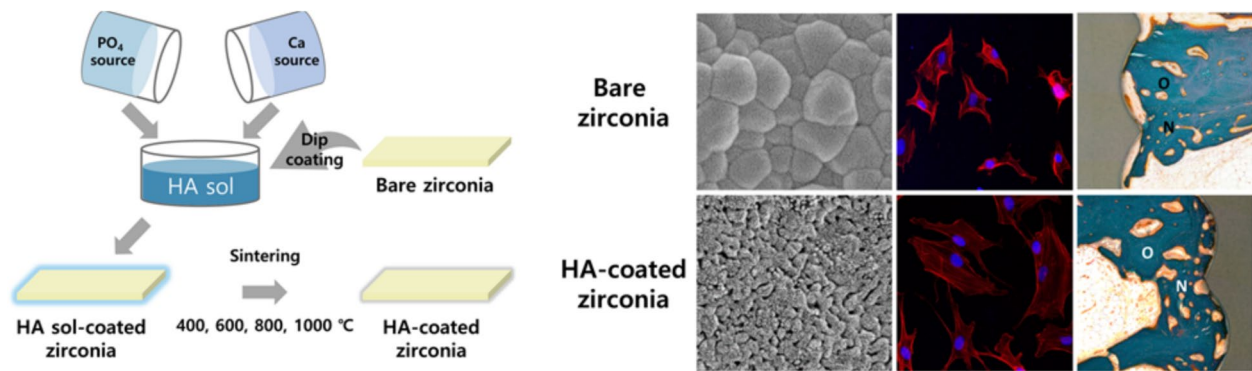


Fig. 15 Comparative analysis of bare and hydroxyapatite-coated zirconia implants. Adapted from reference Kim et al. (2021) with permission

enhances both corrosion protection and mechanical durability. These properties make SiC coatings highly suitable for extending the lifespan of metal components in biomedical and industrial settings.

HAP coating

Hydroxyapatite (HA), which closely resembles the mineral composition of bone, enhances bioactivity and promotes osseointegration. When applied as a coating on zirconia, it transforms the inert surface into a bioactive one, fostering better osseointegration. Moreover, porous zirconia scaffolds coated with HA have been explored as drug delivery systems to aid bone response and ensure effective osseointegration (Cho et al. 2015; Naleway SE et al. 2016; Lavos et al. 2001). Aboushelib et al. (2017) evaluated the osteogenic potential of CAD/CAM porous zirconia scaffolds coated with HA. Their findings revealed a significantly greater new bone formation ($33 \pm 14\%$) on HA-coated scaffolds compared to uncoated ones ($21 \pm 11\%$), suggesting that HA coating enhanced the osteogenic capacity of the zirconia scaffolds. P Karoline et al. (2015) developed coatings by combining Y-TZP and HA in varying proportions. Although higher calcium phosphate content compromised the material's mechanical and chemical stability, it improved its bioactivity upon immersion in simulated body fluid. The study emphasized that coatings with a higher proportion of tetragonal zirconia exhibited superior interfacial bonding, mechanical strength, and bioactivity potential. Calcium phosphate, known for its bioactive properties and its ability to stimulate bone repair, shares a composition similar to biological apatite. However, its coatings are often prone to instability and weak adhesion to the substrate. To overcome these challenges, studies have incorporated tricalcium phosphate-reinforced hydroxyapatite (HA/TCP) coatings on zirconia, which feature pores larger than $100 \mu\text{m}$ and a bonding strength of 24 MPa, ensuring strong interfacial bonding. These coatings also

exhibit absorbable and osteoconductive characteristics (Le Ray et al. 2010; J. Yang et al. 2013).

In their study, Kim et al. (2021) developed a stable hydroxyapatite (HA) coating for zirconia dental implants using a sol–gel technique to enhance osseointegration. Traditional HA coatings often face stability issues due to byproducts produced during high-temperature sintering. To overcome this, the researchers optimized the sol–gel process, resulting in a uniform and stable HA layer on zirconia surfaces (Fig. 15). The implants with this coating exhibited improved biocompatibility and promoted superior bone integration compared to zirconia implants without the coating. This innovative approach holds great potential for improving the performance of zirconia-based dental implants.

Alkali metal coating (magnesium, carbon, and nitrogen)

P Karoline et al. (2015) explored the impact of magnesium (Mg) incorporation into zirconia-calcium phosphate coatings and observed that the addition of Mg promoted enhanced cell proliferation and differentiation when compared to coatings without Mg. Their microstructural analysis revealed a stable, porous coating with robust adhesion. Other research has indicated that magnesium-calcium coatings further enhance the bioactivity of zirconia, with a multilayered hydroxyapatite (HA) precipitate forming a “cauliflower” pattern after 7 days in simulated body fluid (SBF) (Ewais et al. 2017; Schienle et al. 2016). Moreover, introducing elements such as nitrogen and carbon has been shown to improve both the biological and mechanical properties of zirconia. A study highlighted the superior hydrophilicity, bioactivity, and antibacterial characteristics of nitrogen-doped hydrogenated amorphous carbon (a-C:H:N) layers on zirconia. Schienle et al. demonstrated that 3Y-TZP coated with a-C:H:N had significantly reduced bacterial adhesion compared to uncoated zirconia surfaces (Garmendia et al. 2008). Additionally, functionalized multiwalled

carbon nanotube coatings on zirconia led to increased roughness, improved wettability, and greater cell adhesion, all contributing to enhanced osseointegration properties (Kou et al. 2013).

Graphene coating

Li et al. (2014) developed graphene nanosheet (GNS)-reinforced zirconia coatings using plasma spraying. The incorporation of GNS improved the microstructure, hardness, and wear resistance of the zirconia coating. The enhanced properties were mainly attributed to the uniform dispersion of graphene, which contributed to toughening mechanisms like crack deflection and bridging, making the coating more durable for potential wear-resistant applications.

Polydopamine coating

Polydopamine (PDA), derived from the adhesive proteins found in marine mussels, is renowned for its exceptional ability to adhere to both organic and inorganic surfaces, even in underwater environments (Yang et al. 2011). Lee et al. (2007) investigated that the PDA coatings have been applied to a broad range of materials, including noble metals, oxides, semiconductors, synthetic polymers, and ceramics. Research by Liu Y.T. et al. (2013a) has shown that zirconia coated with PDA outperforms uncoated zirconia in promoting cell adhesion, spreading, and proliferation. In addition to its superior adhesive properties, PDA coatings also enhance the antimicrobial properties of zirconia by reducing bacterial adhesion, which supports peri-implant soft tissue regeneration (Liu M. et al. 2015a). Xu et al. (2016) demonstrated that PDA-coated 3D-printed bio-ceramic scaffolds can significantly accelerate osteogenesis, highlighting their potential in bone-related applications. Moreover, PDA has been found to improve the bond strength between resin cement and zirconia, thereby enhancing the overall performance of dental implants (Yu Y et al. 2021). The simplicity and versatility of PDA coatings make them a promising avenue for future research in the modification of zirconia surfaces.

Machined

Bacchelli et al. (2009) examined peri-implant osseointegration and found that zirconia-blasted titanium implants exhibited 35.5% bone-to-implant contact (BIC) after 2 weeks, while machined titanium, titanium plasma-sprayed, and alumina-blasted titanium implants showed 34.5%, 44.7%, and 53.4% BIC, respectively. Their study suggested that zirconia coatings were not superior, likely due to the brief 2-week evaluation period. In a separate study, Gahlert et al. (2007) assessed the removal torque on 64 implants, which included zirconia grit-blasted with

Al₂O₃ particles, machined zirconia, and grit-blasted titanium. Their results indicated higher removal torque for grit-blasted zirconia (40.5 N/cm) and titanium (105.2 N/cm) compared to machined zirconia (25.9 N/cm). Thomas et al. (2015) reported that non-modified zirconia implants demonstrated similar bone-to-implant contact (BIC) and peri-implant soft tissue measurements as acid-etched-sandblasted titanium implants. However, zirconia implants exhibited a higher fracture rate after 6 months of loading. Finally, Hafezeqoran et al. (2017) conducted a systematic review comparing the BIC percentage of machined zirconia and titanium dental implants. The review revealed that the BIC values for machined zirconia (33.74–84.17%) were statistically similar to those for machined titanium implants (31.8–87.95%), concluding that machined zirconia implants could achieve osseointegration on par with machined titanium implants.

Acid etched

Kohal et al. (2004) conducted a study comparing the osseointegration of unloaded zirconia and titanium implants 5 months post-tooth extraction. The titanium implants underwent grit blasting with Al₂O₃ followed by acid etching, whereas the zirconia implants were only grit-blasted. Their findings revealed a peri-implant tissue cuff height of 5 mm around titanium implants and 4.5 mm around zirconia implants. The bone-to-implant contact (BIC) was 72.9% for titanium and 67.4% for zirconia. These results indicated that grit-blasted zirconia implants displayed similar osseointegration and peri-implant soft tissue dimensions as SLA-treated titanium implants. Xie et al. (2015) investigated the impact of three different acid treatments (hydrofluoric acid (HF), acetic acid, and citric acid) on zirconia at various concentrations. They found that HF was the most effective in increasing surface roughness; however, a 40% HF treatment at room temperature notably diminished surface hardness and flexural strength. Their recommendation was to limit HF concentration to 5% to prevent damage to zirconia's mechanical integrity. Fischer et al. (2016) observed a reduction in zirconia strength after 4 h of HF etching, suggesting that 1 h of etching yields optimal results, although it did not significantly enhance the morphological characteristics. Despite HF etching's potential to improve osseointegration, further studies are needed to refine the etching conditions that preserve zirconia's mechanical properties. Roehling et al. (2017) discovered less biofilm growth on grit-blasted and acid-etched zirconia surfaces (ZrO₂-ZLA) compared to grit-blasted titanium surfaces (Ti-SLA). In another study by Flamant et al. (2016), zirconia disks were treated with 5%, 20%, and 40% HF at varying times. The most uniform and rapid roughening occurred with 40% HF for 60 min,

though no strength or fatigue tests were conducted. The etching process dissolved zirconium and yttrium oxide, creating fluoride and hydroxide complexes, with yttrium trifluoride crystals precipitating on the surface. Oh et al. (2017) and Flamant et al. (2016) examined different concentrations and durations of HF etching on pure zirconia, porcelain, and bioactive glass-infiltrated zirconia. They found that neither etching time nor acid concentration had a significant effect on zirconia roughness, while bioactive glass-infiltrated zirconia exhibited a surface comparable to grit-blasted and acid-etched titanium.

Sandblasting

Sandblasting is a technique used to create a rough surface on implants, enhancing osseointegration by improving bone-implant contact (BIC) and removal torque (RTQ), which are indicators of bone growth quality and the strength of the bond between the implant and bone. In a study by Gahlert et al. (2007), zirconia implants with machined and sandblasted surfaces were placed in the maxillary incisor area of thirteen miniature pigs. The results demonstrated that the sandblasted zirconia implants had a rougher surface than the machined ones. Similarly, Yang et al. (2003) explored zirconia coatings, containing 4% CeO₂ and 3% Y₂O₃, applied to titanium and CoCrMo implants via plasma spraying. The surface roughness of these coatings was influenced by the initial powder size and substrate type. The 3% Y₂O₃ zirconia had a powder size of 40–100 µm, while the 4% CeO₂ zirconia had a much finer powder size of 10–20 nm. While no significant differences in hardness were found, the adhesive strength of the 4% CeO₂ zirconia coating on both titanium and CoCrMo substrates exceeded 68 MPa, significantly outperforming the 3% Y₂O₃ coating, which had adhesive strengths of 32.3 MPa on titanium and 24.7 MPa on CoCrMo. In a different study, the effects of machined, sandblasted, and SLA zirconia surfaces were assessed. Both airborne particle abrasion and acid etching increased the surface roughness, with cell proliferation showing notably higher values for surface-treated zirconia at 3 days compared to machined zirconia. However, no significant differences in cell behavior were observed between surface-treated zirconia and SLA titanium after 6 and 12 days (Depprich et al. 2008b, a). Stübinger et al. (2008) studied the effect of various laser irradiations (Er:YAG, CO₂, and diode lasers) on the surface characteristics of polished zirconia implants. SEM analysis revealed no noticeable surface changes with the diode and Er:YAG lasers, while the CO₂ laser caused visible alterations to the zirconia surface. Manzano et al. (2014) conducted a clinical study comparing the performance of zirconia and titanium implants in animal models, focusing on BIC and removal torque. Their analysis

of 16 selected studies revealed no significant differences in BIC and RT between the two materials. The variations observed were attributed to differences in surface treatments and microporosity rather than the materials themselves. The authors concluded that surface-modified zirconia implants could be a viable alternative to titanium implants, though additional clinical studies are needed for further confirmation. Buser et al. (1998) compared the removal torque values (RTVs) of titanium screw-type implants with different surface treatments in miniature pigs' maxillae. Implants with a sandblasted and acid-etched surface (SLA) exhibited significantly higher RTVs than those with a machined and acid-etched surface (Osseotite), indicating stronger bonding at the bone-implant interface for the SLA surface.

In a study, Taniguchi et al. (2016) explored how fiber laser-induced grooves on zirconia implants influenced osteoblast behavior and bone formation. Their findings showed enhanced osteoblast proliferation and differentiation on roughened zirconia surfaces, with increased alkaline phosphatase activity and osteocalcin expression. In vivo tests on rat tibiae demonstrated improved bone-to-implant contact and removal torque for laser-treated zirconia implants, indicating better osseointegration. Aivazi et al. (2018) examined the effects of surface treatments on the viability of L929 fibroblast cells cultured on zirconia nanocomposites. Various treatments, such as sandblasting, acid etching, and laser irradiation, were applied to enhance the surface properties of the nanocomposites. The results, measured via MTT assays, showed that all surface modifications promoted cell proliferation, with laser irradiation leading to the highest cell viability. This suggests that laser treatment optimizes zirconia nanocomposites for better cellular attachment and growth. Lastly, Liu D. et al. (2013b) evaluated three surface treatments—sandblasting, sandblasting followed by regeneration firing, and laser irradiation—on the bond strength between porcelain and zirconia. Both sandblasting and laser irradiation improved shear bond strength (SBS) significantly, with no notable difference between the two methods. X-ray diffraction analysis revealed that sandblasting caused a tetragonal-to-monoclinic phase transformation in zirconia, which was reversed by regeneration firing, while laser irradiation did not alter the crystalline structure of the zirconia. These findings suggest that laser irradiation could be a suitable alternative to sandblasting for improving porcelain-zirconia bonding without affecting the material's crystallographic structure.

Ultraviolet light

T Tuna et al. (2015) explored the impact of UV irradiation on zirconia disks and found that a 15-min UV

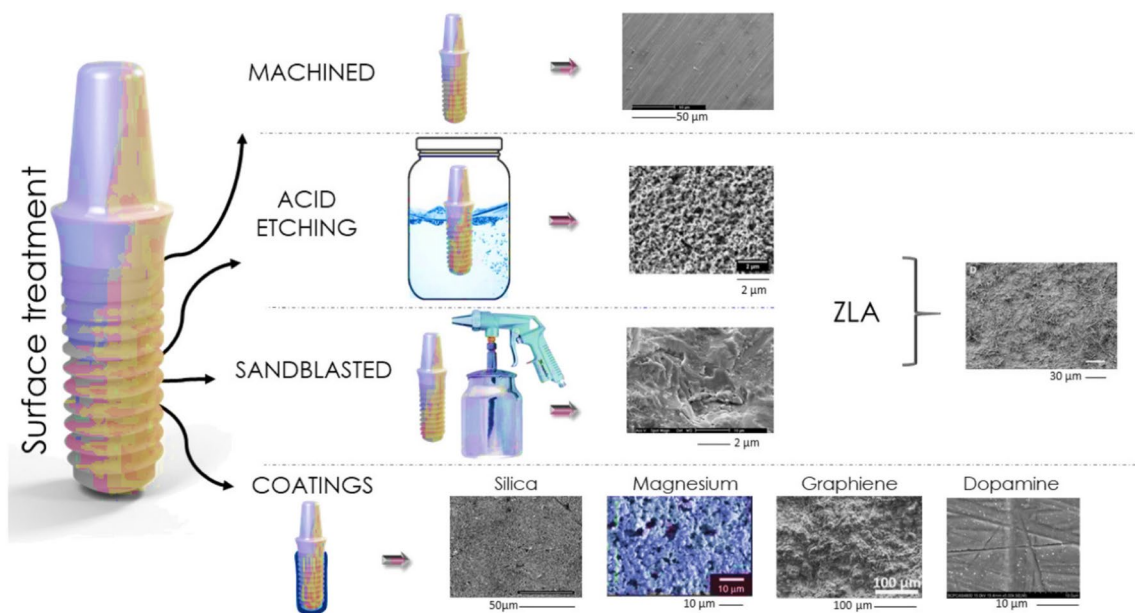


Fig. 16 Schematics and SEM images of various zirconia surface treatments, including machined surfaces (Gahlert et al. 2007), acid-etched surfaces (Flamant et al. 2016), multi-layered coatings (M.S. Laranjeira et al. 2014; G Feng et al. 2021; M. Liu et al. 2015), ZLA-treated surfaces (M. Liu et al. 2015), and grit-blasted surfaces (Liu YT et al. 2013a)

exposure significantly lowered the contact angles, reducing them from 56.4°–69° to 2.5°–14.1°. Their findings revealed notable alterations in the surface composition, including a 43–81% reduction in carbon content and a 19–45% increase in oxygen levels, indicating a transition from hydrophobic to hydrophilic characteristics (Hen-nigesen et al. 2018; Al Qahtani et al. 2015). Additionally, an increase in the monoclinic phase of zirconia was observed. However, other studies did not report a phase transition from tetragonal to monoclinic, suggesting that UV treatment can mitigate zirconia aging without inducing significant crystalline transformations. Further investigation is required to fully comprehend the potential of this approach (Roy M. et al. 2017).

Plant-based coatings

Leveraging plant-based bioactive coatings offers a dual advantage—enhancing microbial resistance and promoting biointegration without compromising biocompatibility. Studies, including those by Wan Mohamad Ikhmal et al. (2020), have shown that such coatings significantly improve the performance of metallic substrates like SS 316, indicating their adaptability to ceramic surfaces. Additionally, plant-derived inhibitors such as *Uncaria cordata*, *Leucas aspera*, and *Salicornia begolovi* have demonstrated high corrosion resistance in acidic environments through adsorption and protective film formation (Selma Lamghafri et al. 2024). Although zirconia inherently resists corrosion, it lacks these organic

interaction mechanisms; thus, integrating plant-based bioactives can introduce antimicrobial functionality and improved bioactivity, aligning with green chemistry and enhancing the clinical longevity of zirconia implants.

Among the various surface treatments explored (Fig. 16), biofunctional coatings—notably hydroxyapatite, polydopamine-mediated mineralization, and phytochemical-based layers—demonstrate superior performance for zirconia dental implants. These modifications promote osteointegration and antimicrobial efficacy while preserving the material's phase and mechanical stability. Advanced nanostructured coatings like graphene further contribute to enhanced cellular responses and durability. In contrast, conventional methods such as sandblasting and acid etching, though effective in roughening, risk tetragonal-to-monoclinic phase transformation. Ultraviolet and machined surfaces, while structurally safe, exhibit limited biological stimulation. Thus, bioactive and multifunctional coating strategies represent the most clinically viable pathway for optimizing zirconia implant integration.

Mechanical strength

Minamizato et al. (1990) investigated the compressive strength of zirconia dental implants with laser-drilled tunnels and found that the implants with tunnels exhibited a reduced compressive strength of 237 kg/mm², compared to 371.5 kg/mm² for those without tunnels. They concluded that zirconia implants, despite the lower

Table 4 Mechanical properties of zirconium and titanium alloy (Apratim et al. 2015)

Mechanical property	Y-TZP	Cp-Ti (grade 2)	Ti-6 Al-4 V alloy
Tensile strength (MPa)	-	345	869 (Annealed)
Bending strength (MPa)	1100 (without HAp) 1800 (with HAp)	100	950
Young modulus (GPa)	210	100	110
Vickers hardness (HV)	1200	150–170	270–320
Fracture toughness (MPa M ^{1/2})	6–8	66	50

strength, retained sufficient mechanical properties for occlusal use. Silva et al. (2009) explored the effect of full crown preparation on the fracture strength of one-piece zirconia implants, reporting fracture strengths of 1023.3 N without preparation and 1111.7 N with preparation. However, other studies indicated that head preparation adversely affected fracture strength. After simulating 5 years of clinical use, zirconia implants fractured at forces between 725 and 850 N without preparation and 539 and 607 N with preparation, indicating that zirconia implants maintain adequate fracture strength within clinical limits. The mechanical characteristics of Y-TZP, CP Ti, and Ti-6 Al-4 V alloys for dental implants vary significantly as shown in Table 4. Zirconia (Y-TZP) has a bending strength of 1000 MPa, which is appropriate for dental applications, though it is less robust than titanium alloys. Its fracture toughness, ranging from 6 to 8 MPa m^{1/2}, is higher than alumina but considerably lower than titanium (66 MPa m^{1/2}). Zirconia also has lower notch stability, which diminishes its fracture resistance compared to titanium. Due to its high Young's modulus of 210 GPa, zirconia is less efficient at absorbing occlusal forces than titanium alloys, which have a modulus between 100 and 110 GPa. This stiffness may also lead to problems with screw fixation, resulting in loosening (Apratim et al. 2015). Qeblawi et al. (2010) examined the effect of various mechanical surface treatments on the flexural strength of yttria-partially stabilized zirconia (4 × 5 × 40 mm), including (1) control (no treatment); (2) airborne-particle abrasion (APA); (3) silicon coating; and (4) wet hand grinding. The results showed the following flexural strengths in MPa: control (571.7 ± 79.2), APA (798.8 ± 138.2), silicoated (594.3 ± 100.5), and hand ground (1727.7 ± 112.7), with hand grinding yielding the highest strength.

Abrasion resistance and material durability

An ideal dental restoration should exhibit excellent wear resistance without causing excessive wear to the opposing natural teeth. Zirconia, a rigid ceramic material, stands out for its superior wear resistance when compared to both human enamel and porcelain. Research has investigated whether zirconia, especially non-veneered zirconia, leads to significant wear on opposing teeth. Oh et al. (2002) explored the factors influencing enamel abrasion caused by dental ceramics and concluded that there is no direct correlation between ceramic hardness and enamel wear. Instead, the microstructure, surface roughness, and individual patient risk factors have a more substantial impact. For full-contour zirconia crowns, appropriate surface treatment is essential. Beuer et al. (2012) demonstrated that glazing helps maintain zirconia's wear resistance, preventing excessive wear on opposing teeth. However, the 30–50-μm glazed layer tends to wear off after approximately 6 months, revealing a rougher surface that could increase abrasion. Polishing zirconia to a smooth finish, similar to glazing, further enhances wear resistance. Despite this, full-contour zirconia polished with 3-μm diamond paste still does not fully protect enamel from excessive abrasion.

Long-term reliability

Flinn et al. (2012) examined the aging resistance of three commercial 3Y-TZP zirconia grades: Lava, Zirkon Zahn, and Zirprime. After exposing the materials to 200 h of autoclaving at 134 °C and 0.2 MPa, they found that the monoclinic phase content varied between 25 and 80%, with the depth of the transformed zone ranging from 2 to 60 μm. The biaxial flexural strength was measured between 100 and 700 MPa. The study highlighted that the zirconia's resistance to aging is influenced by factors such as the material's initial properties, processing methods, and microstructure. It was also observed that reducing the grain size of zirconia could lower the driving force for phase transformation, thus stabilizing the tetragonal phase and enhancing the material's aging resistance. In a critical review, Lughi et al. (2010) examined low-temperature degradation (LTD), a key factor in the aging of zirconia ceramics for dental use. They emphasized that LTD is primarily governed by factors such as stabilizer type, concentration, grain size, and residual stresses. The authors warned against using accelerated aging tests to predict the long-term performance of zirconia, as these tests may not accurately simulate real-life conditions. They suggested optimizing stabilizer content and controlling grain size to improve the durability and longevity of zirconia dental restorations. Hallmann et al. (2012) found that zirconia ceramics sintered at temperatures between 1350 and 1450 °C exhibited improved

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Fig. 17 Advancement in material and technology in ceramic dental implants from the 1980 s to the 2000 s. Adapted from reference Yuriy May et al. (2025)

aging resistance, with finer, more uniform grains and better stress distribution. As the sintering temperature increased, both the tensile/compressive stress ratio and monoclinic phase content also increased. Research on doping with elements like Al, Ce, and Si showed that these dopants could help mitigate LTD by enhancing the material's stability. Ardlin et al. (2002) noted that the aging behavior of colored zirconia has been less studied, but their findings indicated that although the monoclinic phase content increased slightly after aging, the bending strength of the material remained largely unchanged, suggesting that aging does not significantly compromise the material's strength in dental applications (Samudurova et al. 2015). Nakamura et al. (2015) examined the impact of LTD and cyclic loading on the fracture resistance of monolithic zirconia molar crowns. After autoclaving zirconia samples at 134 °C for various durations, they observed an increase in the monoclinic phase fraction with longer autoclaving times, which plateaued after 50 h. The depth of the monoclinic phase continued to increase, but the fracture resistance decreased from 5683 to 3975 N after 100 h. Despite this reduction in strength, the crowns still maintained adequate fracture resistance for typical chewing forces in the molar region. Zirconia, being an inert ceramic, does not naturally bond well with other materials, so surface modifications are essential to enhance adhesion. Y-TZP surfaces are typically treated through mechanical methods such as sandblasting with Al_2O_3 or tribochemical silica coating, followed by chemical treatment with primers like MDP. The phosphate groups in MDP promote bonding with the zirconia surface, improving adhesion. However, these treatments

may sometimes induce micro-cracks or phase transformations, potentially affecting the long-term performance of the zirconia material (Cheung GC. et al. (2014).

Progression of zirconia ceramic implants

Dental implantology in Europe saw pioneering advancements (Fig. 17) in both titanium and ceramic implants. Swedish inventor Per-Ingvar Branemark created the first titanium implant in the 1960 s, while ceramic implants emerged as an alternative due to concerns about metal's potential immunological, aesthetic, and allergic issues. Ceramic implants, particularly made from zirconia, have been in clinical use since the late 1960 s. Zirconia implants were first developed in 1967 by Swiss dentist Sami Sandhaus, who believed zirconia's tooth-like color, biocompatibility, and mechanical properties made it an ideal implant material. By the 1970 s, alumina ceramic implants were also being tested, with research showing their high bio-inertness, aesthetic appeal, and ability to integrate with bone. In the 1980 s, zirconia-toughened alumina (ZTA) implants replaced alumina, leading to the development of today's zirconia implants made from yttria-stabilized zirconia (Y-TZP), which offers superior strength and resilience compared to earlier materials. Despite early setbacks in acceptance, ceramic implants gained recognition by the 1980 s, especially after Sami Sandhaus reported positive clinical outcomes in 1997. Today, zirconia implants remain a strong alternative to titanium, offering advantages in aesthetic appearance, biocompatibility, and reduced plaque buildup (Yuriy May et al. 2025).

Summary and conclusion

Zirconia (ZrO_2) stands out in orthopedic applications for its superior mechanical strength, fracture toughness, and wear resistance. Its excellent biocompatibility and corrosion resistance make it ideal for load-bearing implants, coatings, and bone scaffolds. Advances in composites like yttria-stabilized zirconia (YSZ) and zirconia-toughened alumina have further enhanced its performance. To improve osseointegration, surface treatments such as laser texturing, biofunctionalization, and acid etching are employed, optimizing surface chemistry and topography. Overall, zirconia's robust properties and adaptability through surface engineering make it a premier choice for next-generation orthopedic implants.

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Authors' contributions

Gopi Srinivasan: conceptualization, investigation, writing—original draft. Anushiya Manickam: investigation and methodology, software. Sivasakthi Sivakumar: data curation and methodology. Jeevadarani Murugan: data curation, resources. Shinyjoy Elangomannan: investigation, formal analysis. M. Surendiran: supervision, project administration.

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Data availability

Data will be made available on request.

Declarations

Ethics approval and consent to participate

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Competing interests

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